Chapter 23

Clinical Trials

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Learning Objectives:

Objective 23.1 Introduce clinical research methods.
Objective 23.2 Detail methodological aspects in clinical trials.
Objective 23.3 Explain the phases of clinical trials.
Objective 23.4 Discuss the role of pharmacists and newer technologies in clinical research.

OBJECTIVE 23.1. INTRODUCE CLINICAL RESEARCH METHODS

Clinical research involves all research in humans with different study designs. [Clinical means humans and Trials means Experiment][1,2] Clinical research is synonymously used for clinical trials. A clinical trial is any research study that prospectively assigns human participants or groups of humans to one or more health-related interventions to evaluate the effects on predefined health outcomes.[3] As per regulatory requirements, clinical trials are mandatory before drugs are approved and marketed.[4]

First Controlled Clinical Research

There are many records in history for the conduct of studies in humans. In the 18th century (early modern era), one of the first controlled studies was performed by a Scottish physician named James Lind, who is regarded as the "Father of Naval Medicine." Scurvy was considered a dreadful disease among sailors. Sailors with scurvy (gum disease, dry mouth, dry eyes, loss of teeth) were provided with citrus fruits such as oranges and lemons, cider, sulfuric acid, vinegar, salt water and garlic paste, mustard seed, horseradish, balsam of Peru, and gum myrrh. The sailors supplemented with citrus fruits recovered. This was the first controlled study performed across sailors for the treatment of scurvy.[5,6]

Use of Placebo

In the late 19th century, the term placebo was introduced in the field of medicine. In 1863, Austin Flint, a physician, prescribed a placebo (noneffective or fake drug) for his patients with rheumatic fever. The placebo used was a largely diluted tincture of Quassia. He concluded that in more than 90% of patients treated with placebo, there were no significant differences in the results when compared with the ones with active treatment group. Many physicians started with placebo treatment (mint water) for their patients.[5,6]

Use of Randomization

In 1946, 107 patients with acute pulmonary tuberculosis were involved in a clinical trial held in the United Kingdom.[7] As the supply of streptomycin was limited, the patients were randomly selected and assigned to one of the treatment groups. The random selection was indeed blinded to the investigators and coinvestigators.

- Group I—55 patients treated with bed rest plus streptomycin (the active treatment group)
- Group II—52 patients treated with bed rest alone (the control group)

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The observational period for the trial was 6 months. Numbers were randomly assigned to each sex at each center in a closed envelope. The patients on admission were randomized to the groups based on the number on the envelope.[4–6]

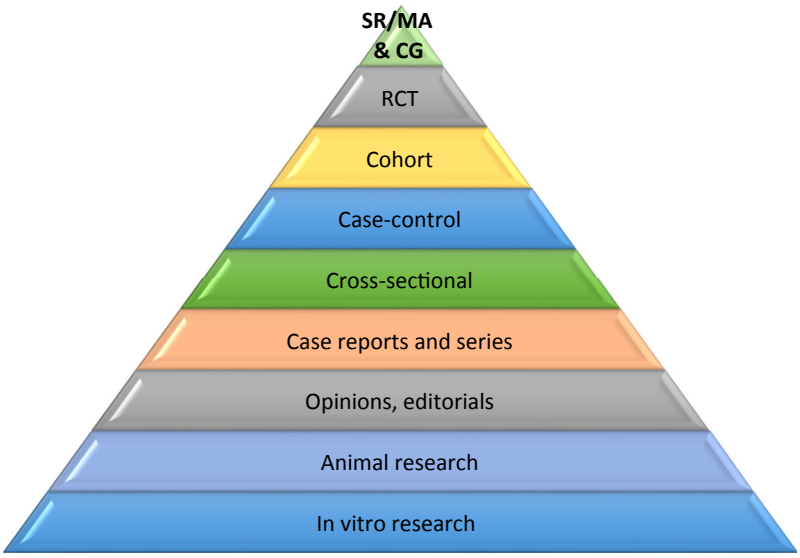
In the 20th and 21st centuries, various predetermined randomization techniques for appropriate outcomes are available. In addition, the phases of clinical trials are also developing and may include a larger population for the research purpose of achieving scientific results.

Clinical Research Methods

One of the key objectives of clinical research is to generate health-related information. Randomized clinical trials are considered as the gold standard in generating high-quality medical information, but epidemiologists challenge that clinical trials do not generate real-world data. Studies at real-practice settings with natural bias mimic the results that are normally achieved once a drug is marketed (more realistic outcomes). Pragmatic trials solve this issue by being conducting in real practice settings.[8] The differences of clinical trials and observational studies are compared in Table 23.1, and types of research methods are shown in Figs. 23.1 and 23.2.

TABLE 23.1 Differences Between Clinical Trials and Observational Studies

Clinical Trials	Observational Studies
Experimental	Observational
Data from controlled settings, which is difficult to achieve in real practice	Real-world data are natural confounders
Expensive	Relatively less expensive, retrospective studies are even cheaper than cohort studies
Long duration	Relatively shorter duration when compared to clinical trials
Could be multicentered	Possible in any facility
Mostly sponsored by pharmaceutical companies	Mostly performed by independent agencies, professionals
Populations are covered, but limited in many aspects like pregnant, pediatric, geriatric, etc.	Flexible to perform in different populations
Unethical to study mutagenicity studies	Less ethical concerns as interventions are not required



- SR/MA: Systematic review/metaanalysis
- CG: Clinical guidelines
- RCT: Randomized control trial

FIGURE 23.1 Types of research based on quality of evidence generated with higher quality research methods at the top.

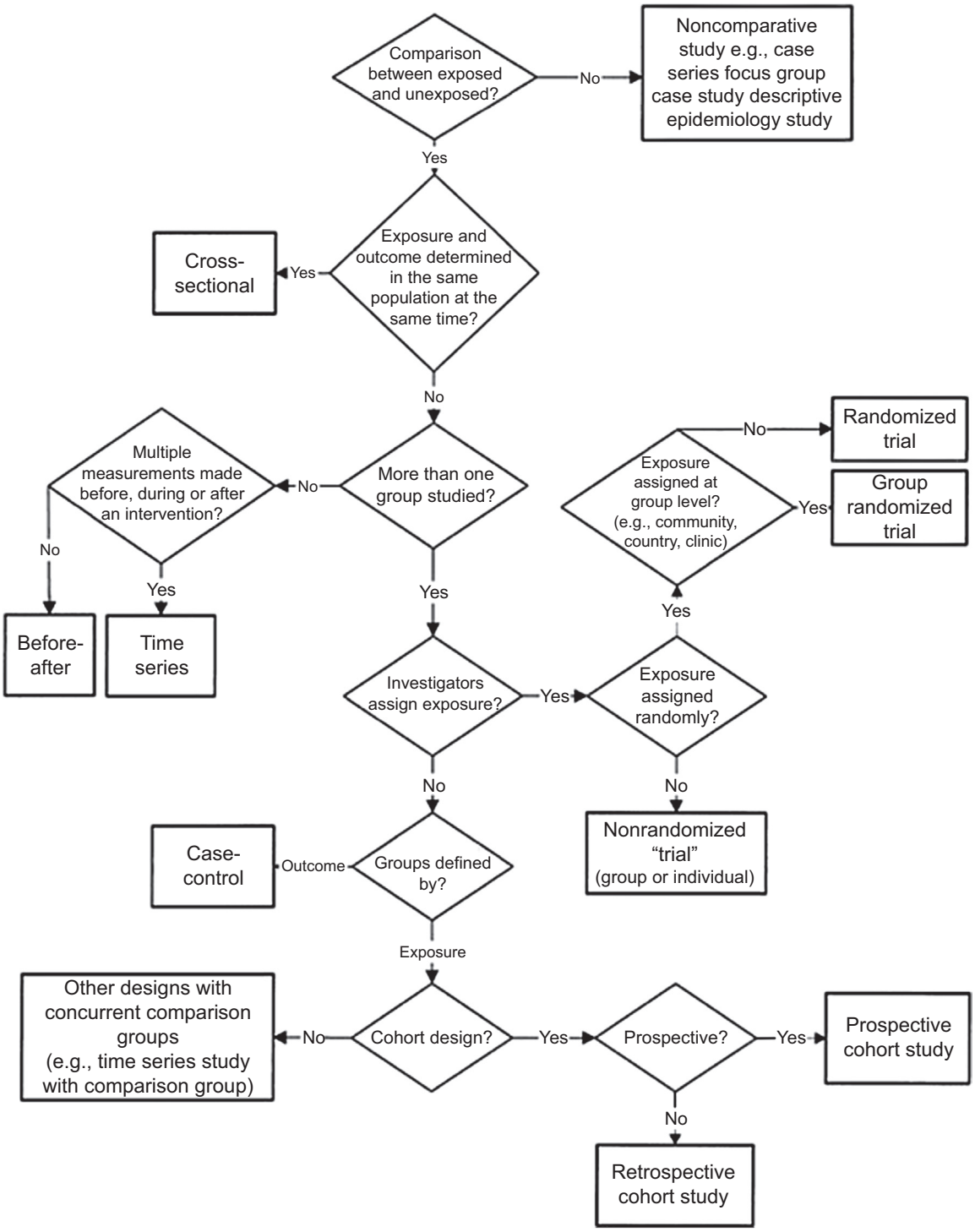


FIGURE 23.2 Choice of different clinical research methods.

OBJECTIVE 23.2. DETAIL METHODOLOGICAL ASPECTS IN CLINICAL TRIALS

Preclinical studies are conducted in lab animals or by other laboratory techniques while clinical trials are conducted in human beings. There are two types of preclinical research: in vitro (the experiment is done in glass or plastic vessels in the laboratory) and in vivo (the experiment is done in the body of a living organism). Preclinical studies are not very large in size but can provide detailed information on dosing and toxicity levels, respecting Good Laboratory Practice by an approved written protocol that clearly indicates the objectives and all methods of how to conduct the study (Fig. 23.3).[15]

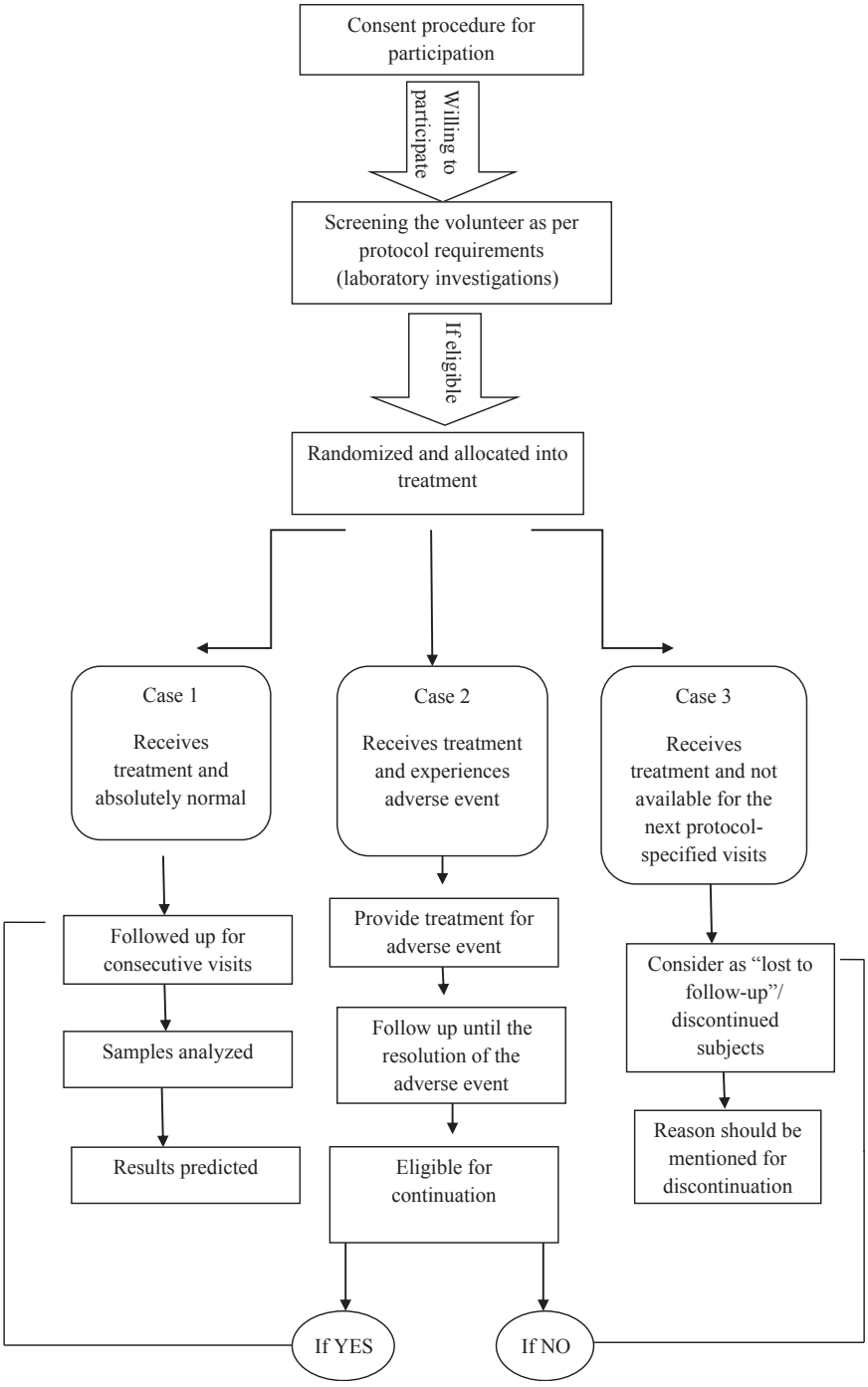


FIGURE 23.3 Clinical trial activity chart.

Randomization

Randomization is used in clinical trials to reduce the possibility of bias. Bias here means the systematic tendency of any aspects of the design, conduct, analysis, and interpretation of the results of clinical trials to estimate a treatment effect deviating from its true value.[7] Therefore, randomized trials are performed to reduce potential bias. Randomization can be generated using various statistical methods. There are various types of randomization: simple randomization, block randomization, stratified randomization, adaptive randomization, and minimization methods are detailed as follows (Fig. 23.4).[9–12]

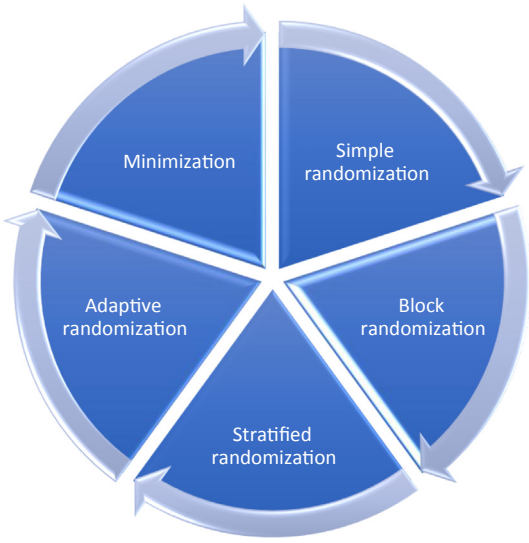


FIGURE 23.4 Types of randomization.

Simple Randomization

In this type, patients/subjects are allocated randomly to different treatment groups. As a simple strategy, this type of randomization could be performed by flipping a coin, or by throwing a dice, or even by shuffling cards. In large clinical research simple randomization generates a similar number of participants among groups, but in a small sample size, this method is problamatic. An example is given in Table 23.2; out of 15 patients, 5 of them randomized into group A and 10 randomized into group B. As there is no actual balance in numbers, the outcome variables are therefore imbalanced.[11]

Block Randomization

Here, the sample size or participants are equally distributed all the time using blocks. Blocks are small and balanced with predetermined group assignments. The block size is determined by the researcher and should be a multiple of the number of groups (i.e., with two treatment groups, block size of either 4, 6, or 8). Blocks are best used in smaller increments as researchers can more easily control balance.[11,13]

An example: 2 blocks with 2 treatment groups.
Treatment 1—Control (C); Treatment 2—Treatment (T).
Block 1—CT; Block 2—TC (Table 23.3).
Similarly, 4 treatment groups with 4 blocks can be assigned as randomized in Table 23.4.
T1—A; T2—B; T3—C; T4—D.
B1—ABCD; B2—BCDA; B3—CDAB; B4—DABC.

TABLE 23.2 Example of Simple Randomization

Group A	Group B
1	2
4	3
8	5
11	6
12	7
	9
	10
	13
	14
	15

TABLE 23.3 Example of Block Randomization—Two Blocks

Block	Subject No.	Group A	Group B
B1	1	C	T
B2	2	T	C
B1	3	C	T
B2	4	T	C
B2	5	T	C
B2	6	T	C
B1	7	C	T
B2	8	T	C
B1	9	C	T
B1	10	C	T
B2	11	T	C
B2	12	T	C
B1	13	C	T
B1	14	C	T
B2	15	T	C
B1	16	C	T
B2	17	T	C
B1	18	C	T
B1	19	C	T
B2	20	T	C

The groups generated are usually not comparable for the relevant comorbidities. This reduces the power of the study though a balance in sample size is achieved, and hence stratified randomization is preferred.

Stratified Randomization

Strata means group. Variables such as age, race, gender, and underlying disease (comorbidities) conditions are generally factorized/stratified. In this method, the participants are first divided into strata and then block randomization is applied. A separate block for each combination of covariates is to be generated (Fig. 23.5).

Having many covatiates to stratify complicates randomization. In addition, practically it is difficult to enroll all participants at once to clinical trials, enrollment in trials keeps happening over a period of time. Thus it is difficult to stratify potential participants to different strata before generative blocks. Adaptive randomization has the benefit of enrolling participants one by one with a balance of covariates.[11]

Adaptive Randomization

As the name suggests treatment is adapted depending on the previously assigned treatment. It is also termed "Response Adapted Randomization." This type of randomization is certainly drawing a ball from a basket. The balls of two treatments, say "x" or "y," are chosen on randomization. The investigator decides the treatment (test vs. control) as the outcome would be known immediately. If patient "A" gets "x" treatment and survives and the next patient "B" gets "y" treatment and dies, the consecutive patient would automatically receive "x" treatment. Thus the number of "x" balls would be added to the basket, and this continues until a failure is reached in "x" treatment. Generally, a ratio status of 1:1 (x:y) would be maintained in the beginning and as the treatment continues the ratio might change (3:1). This indicates that the "x" treatment has been carried out in three subjects successfully. However, the decision to choose the treatment may not be possible in a larger sample size study as this method is sometimes disagreeable so is not very often used.[14]

TABLE 23.4 Example of Block Randomization—Four Blocks

Block	Subject No.	Group A	Group B	Group C	Group D
B1	1	A	B	C	D
B2	2	B	C	D	A
B3	3	C	D	A	B
B4	4	D	A	B	C
B1	5	A	B	C	D
B3	6	C	D	A	B
B3	7	C	D	A	B
B2	8	B	C	D	A
B4	9	D	A	B	C
B1	10	A	B	C	D
B1	11	A	B	C	D
B2	12	B	C	D	A
B4	13	D	A	B	C
B2	14	B	C	D	A
B3	15	C	D	A	B
B4	16	D	A	B	C
B1	17	A	B	C	D
B1	18	A	B	C	D
B2	19	B	C	D	A
B3	20	C	D	A	B
B4	21	D	A	B	C
B2	22	B	C	D	A
B3	23	C	D	A	B
B1	24	A	B	C	D
B4	25	D	A	B	C
B1	26	A	B	C	D
B2	27	B	C	D	A
B3	28	C	D	A	B
B4	29	D	A	B	C
B3	30	C	D	A	B
B4	31	D	A	B	C
B4	32	D	A	B	C
B1	33	A	B	C	D
B2	34	B	C	D	A
B1	35	A	B	C	D
B2	36	B	C	D	A
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B2	39	B	C	D	A
B3	40	C	D	A	B

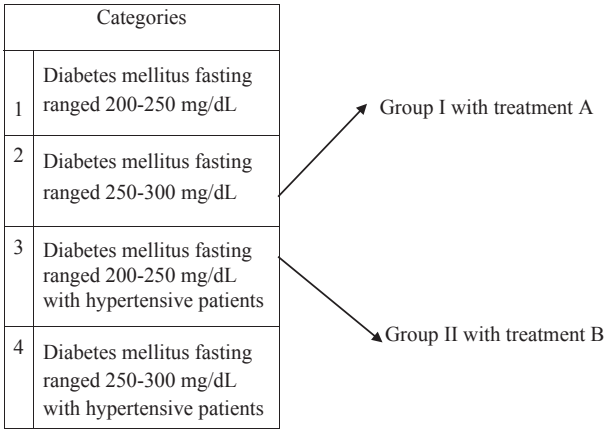


FIGURE 23.5 Example of stratifying covariates in stratified randomization.

Minimization

Minimization is a rather complicated method of adaptive randomization. This is a method that balances the number of patients throughout a stratum. As discussed in "Stratified Randomization" the number of individuals in each stratum are summed up, and the next patient is assigned to a lesser enrolled group. A pattern of this randomization technique is shown in Table 23.5A. Signs of difference as shown in Table 23.5B help in the allocation of the next patients.

A sign of the difference between the two treatments is introduced. Considering the total treatment X has more numbers, allocate the next participant to Y. Also considering the sign of differences, treatment X is greater. Allocate the next

TABLE 23.5A Example of Minimization

Strata	Treatment X	Treatment Y
Sex		
Male	10	8
Female	5	5
Age		
50–64 years	4	6
65 years and above	11	7
Colorectal Cancer		
Stage-1	7	4
Stage-2	2	3
Stage-3	4	4
Stage-4	2	2
Total	15	13

TABLE 23.5B Signs of difference in minimization technique

Strata	Treatment X	Treatment Y	Sign of Difference
Sex			
Male	10	8	+
Age			
65 years and above	11	7	+
Colorectal cancer			
Stage-2	2	3	–
Total	23	18	

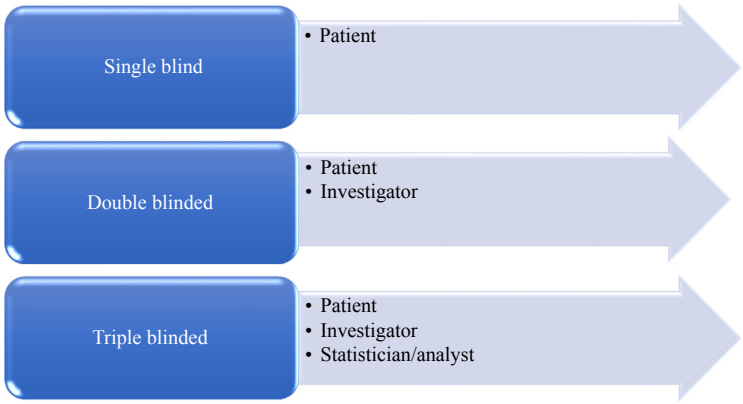


FIGURE 23.6 Types of Blinding.

participant to Y. Such type of randomization can allocate the numbers participating in each treatment group by considering the balance in prognostic factors too. In case if there is an equal allocation between the treatment numbers (e.g., there is a balance in the prognostic variables), mere randomization can be followed. Allocate the next subject in case of such a tie situation to treatment X with a higher assigned probability.

Blinding

The trials are blinded to eliminate bias as well. The tendency for such bias to occur may be unintentional. Nevertheless, blinding should be considered. If blinding is planned, it should be described during the stages of preparation of the protocol. Open label or nonblinded trials are generally adopted in bioequivalence (BE) studies that are conducted to bring generic drugs to the market. In single blinding just the patient is unaware of what they are receiving (investigational product, standard treatment or placebo). In some cases if there are understandable differences between investigational product and standard treatment (e.g. once daily and twice daily dosing), the investigator could figure out which product is used and could not be blinded. In this case the statistician/analyst could be blinded (Fig. 23.6).[10]

Controls

There are different types of control groups predicted for each type of clinical trial.[9]

1. Placebo concurrent control: In this way, a group of people receives a placebo (which does not contain the test drug) and are compared with a test formulation. The fake or dummy drug (placebo) may substantially improve the health condition of the subjects psychologically (he or she realizes the improvement is because of taking a drug). Such an effect is called a "placebo effect."
2. No-treatment concurrent control: In this way, the patients are observed without providing any treatment. However, their outcome will be compared against the interventional test outcome and is thoroughly applicable for nonblinded studies. Study end points should be objective.
3. Dose–response concurrent control: Doses may be fixed and gradually increased. Two such differences in doses (different groups) are compared for a dose–response relationship.
4. Active (positive) concurrent control: Positive control group includes people who are tested with a drug that would give a known response and an expected outcome as that of the test formulation.
5. External (historical) control: The control refers to a group of people who are treated with the same test intervention at the same time but in a different site (external). It can also be historical control, which includes and compares a group of patients treated at an earlier time.[9]

STUDY DESIGNS

Crossover Studies

In crossover studies, the study participants will be switched throughout to all the treatment groups (both test and reference formulations) after a washout period. Being the same set of the population the advantage of crossover studies is that patients act as their own controls. Bioavailability (BA)/BE studies are usually conducted as crossover studies. BE studies can be conducted under fasting and fed conditions. The sampling time points and the duration of BE studies depend on the half-life of the drug of interest. In the case of a feeding study, a high-fat, high-calorie breakfast would be provided to the subjects before administration of investigational product (IP).

Parallel Studies

For drugs with long half-lives where an adequate washout period cannot be given between two treatments, it is preferred to conduct a parallel design instead of a crossover design. In this design, participants do not receive both test and reference formulations. They receive one of the formulations as per the randomization. Usually, this would be a single period study. However, randomization would be generated to know which participant receives a test or a reference.

Steady-State Studies

In steady-state studies, subjects are given multiple doses of IPs so that a steady state can be attained before evaluation of the pharmacokinetic profile of the drug is conducted.

Basket and Umbrella Trials

Clinical trial designs are evolving as a response to precision medicine; the traditional types of clinical trials are not viable in some cases (for example, lung cancer). Over the past several years two clinical trial designs have been proposed and implemented in many oncology trials: basket (or bucket) trials and umbrella trials. The basket and umbrella trial designs avoid overtreatment and save valuable resources by matching the right drug to the right subgroup of patients through genetic biomarkers. Although basket and umbrella trial designs are almost exclusively in oncology trials, we can assume that such designs can be generalized to other therapeutic areas as well. A basket trial is especially meaningful for "rare" diseases, where subjects are difficult to find. Biomarkers are used to identify patients likely to respond to treatment.[27] For example, the umbrella patients with advanced nonsmall cell lung cancer enrolled in an umbrella trial undergo a tumor biomarker analysis that is used to assign them to a phase II study. The differences between these two types of clinical trials are:

- An umbrella trial researches one disease/one cancer type, multiple subgroups identified by the biomarkers, many drugs—many patient groups/arms; a basket/a bucket trial researches many diseases/many cancer types, a single subgroup, one drug.
- An umbrella trial defines cancer as a body location or histology (for example, breast cancer, lung cancer, etc.); a basket trial defines cancer based on the genetic aberrant or biomarker signature (for example, cancer with X-positive biomarker).[28] An umbrella trial allows the researcher to test different drugs and patients with the same disease family and identifies those who respond best to a specific therapy; a bucket trial understands the differences between different types of cancers—all contain the mutation in question.[29]

Adaptive Study Design

In traditional clinical studies, groups of patients receive a predetermined therapy for a fixed period. In an adaptive trial, researchers can see how patients involved in the study are doing while the trial is still running; various aspects of the study design can be changed on the fly or even react to new research insights that happened during the study. New clinical study designs and protocols like these help to transform advances in cancer research into improved options for patients as fast as possible in an effective way.[30]

Registry Trials

Randomized clinical trials provide a foundation of clinical evidence; but they have limitations, including increasingly prohibitive costs of conducting adequately powered studies. Registry trials, which are observational studies, can be used efficiently and effectively to assess hard clinical end points in large patient cohorts; this study design is well suited to hospital settings and registry networks. Despite these advantages, registry studies have disadvantages as well: data quality can be variable and questionable, comparative outcomes research is questionable, and adjustment for possible confounding factors is needed. Potential confounding variables could be removed through randomization in registry-based randomized clinical trials.[31]

OBJECTIVE 23.3. EXPLAIN THE PHASES OF CLINICAL TRIALS

Phases of Clinical Trials

There are five different phases of clinical trials. Clinical trials will start once the preclinical studies give promising results. The five phases of clinical trials are as follows (Fig. 23.7)[16,17]:

- Phase 0
- Phase 1



FIGURE 23.7 Phases of clinical trials.

- Phase 2
- Phase 3
- Phase 4

Phase 0

Phase 0 studies are not common. These studies are called microdosing studies. This phase is usually carried with fewer subjects (generally healthy subjects). The phase ascertains the preliminary pharmacokinetic properties of the drug (absorption, distribution, metabolism, and excretion—ADME). It is very difficult to establish the toxicity criteria with such microdosing phenomena. The outcome of the study would rely on or would be compared with preclinical studies done on animals. However, a therapeutic benefit cannot be determined at this phase. Usually, phase 0 studies will be conducted to avoid exposure of the IP to a higher number of subjects in phase 1. Phase 0 studies will provide the preliminary pharmacokinetic information of the lead compound. Microdosing will help in the drug development process without adversely affecting the safety of the subjects.[18]

Phase 1

Usually carried out in a smaller number of people (12−30 numbers) and performed after animal toxicity studies or phase 0 studies are completed. This phase has been carried out with healthy volunteers with exceptions being patients who are being treated for a disease such as cancer and HIV/AIDS. As it is with fewer people, the researchers can monitor the maximum dose required with fewer side effects. These are also called dose escalation studies. Single and multiple doses will be escalated gradually to determine the maximum tolerated dose (MTD).

Safety and tolerability of the drug are determined at this level.

Phase 1 studies are executed in two methodologies:

- Single ascending dose study—SAD
- Multiple ascending dose study—MAD

Single Ascending Dose Study

In a given period of time (e.g., up to 3 months), pharmacokinetic parameters are determined. Three to four volunteers (a small group) receive the lowest single dose (e.g., 1 mg) and are observed for any toxicity/adverse events. If no adverse events and toxicity are established at this dose level of 1 mg, this type of study can be escalated sequentially to a new small group of volunteers with the next higher dose (2 mg). If toxicity is observed at a 1 mg dose, however, the same tested dose (1 mg) is administered to a new set of three subjects and observed for adverse events. This is done until the volunteers experience an adverse event at a defined dose, and the dose before this is considered as the MTD.[17]

Maximum Tolerated Dose

MTD refers to the highest dose of a pharmacological drug, which would produce the desired effect without unacceptable toxicity.[13]

Example of Single Ascending Dose

Considering an X axis as time in hours (h) and a Y axis as the average of volunteers plasma drug concentration in nanogram/milliliter (ng/mL) (Fig. 23.8).

The graph depicts that 1 mg (group I) of drug X has been initiated in a small group of people (in 3−5 numbers). The pharmacokinetic data are predicted. With no toxicity is established at this level, the dose of drug X is further escalated to 2 mg (group II) and 3 mg (group III) with a different group of people. The dose escalation continues as long as the maximum concentration is reached as stated in the protocol. The dose of 4 mg (group IV) is discontinued before the concentration maximum could be reached due to the possible occurrence of unacceptable toxicity. Hence MTD is decided as 3 mg for drug X. Further dose escalation is terminated (Fig. 23.9).

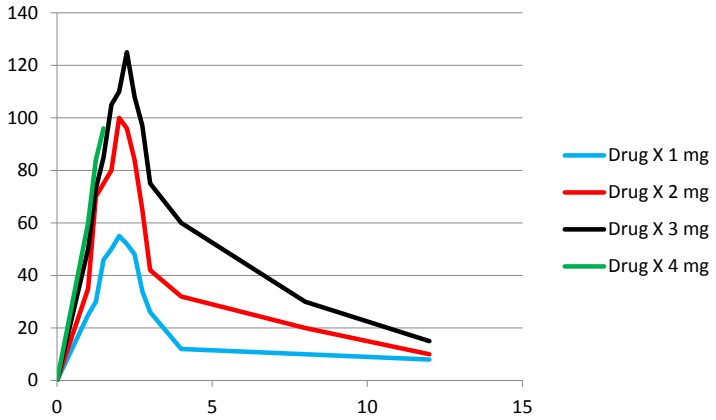


FIGURE 23.8 Example of single ascending dose (Plasma drug concentration on Y-axis and time on X-axis).

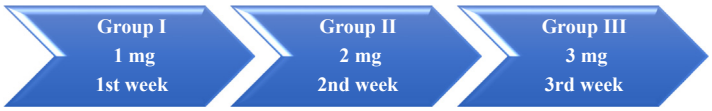


FIGURE 23.9 Phase 1—single ascending dose study.

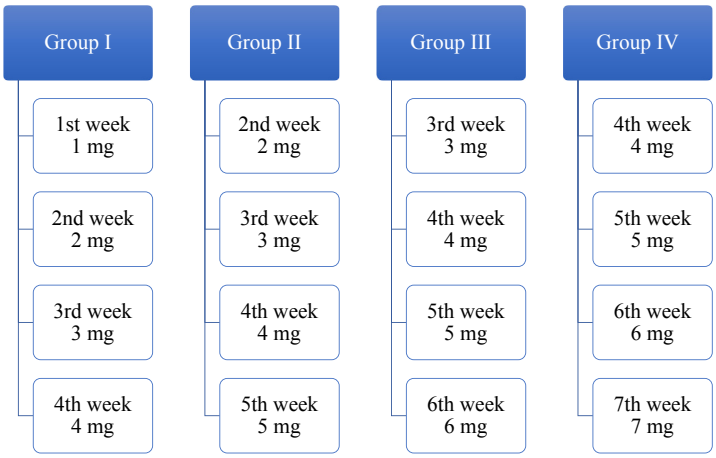


FIGURE 23.10 Phase 1—multiple ascending dose study.

Multiple Ascending Dose Study

A group of people receives multiple doses of a drug to determine pharmacokinetic and pharmacodynamic properties. A second group receives an escalated amount of multiple doses, and the same would be determined at various groups until tolerability is established (Fig. 23.10).[19]

From the figure, dose escalation happens within the group to study the pharmacokinetic parameters. The fate of the drug is determined with multiple dose levels. The escalation of the dose levels among different groups is certainly decided from SAD studies to evaluate the therapeutic drug concentration in the human body.

Example of Multiple Ascending Dose

According to the graphs (X axis: time in hours and Y axis: concentration in ng/mL), the concentration maximum (Cmax) lies around 55 ng/mL (Fig. 23.11A–G) to 180 ng/mL (Fig. 23.11A–G) and the time taken to reach the maximum concentration is 0–4 h. Assessment of the pharmacokinetics parameter using multiple groups assists in determining the toxicity profile.

Phase 2—Therapeutic Exploratory Phase

This phase handles a considerably greater number of patients compared to phase I. The numbers may gradually increase to 100 or more depending on the number of sites. Unlike phase I trials, these trials are intended to treat the patients with an expected indication to find out safety issues as the primary objective. Therefore this is also termed a therapeutic exploratory phase. Safety of the drug is assessed with priority followed by its efficacy.[19] This may be categorized as phase 2a and phase 2b.

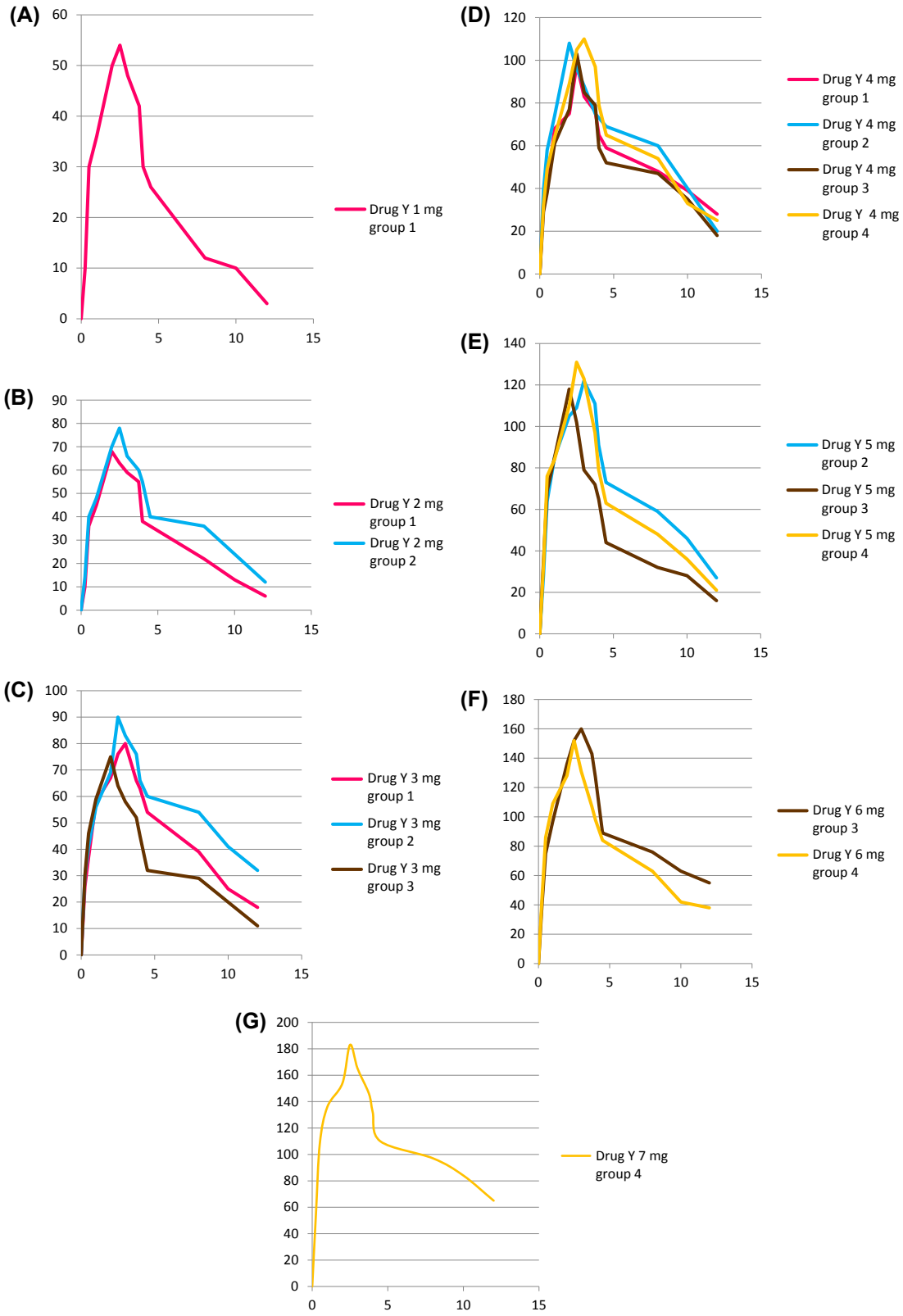


FIGURE 23.11 (A–G) Examples of multiple ascending dose (Plasma drug concentration on Y-axis and time on X-axis).

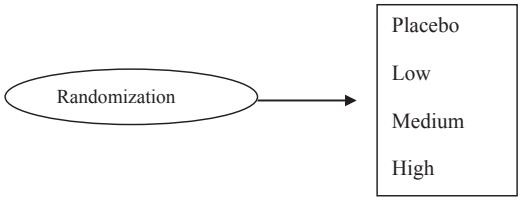


FIGURE 23.12 Proof of concept studies—randomization.

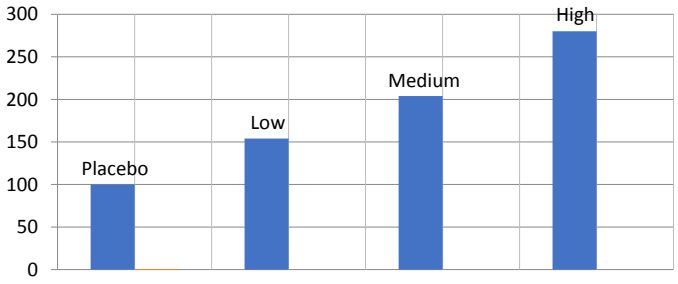


FIGURE 23.13 Responses for different doses.

Phase 2a—Proof of Clinical Concept Studies

These studies are pilot studies, single centric, and a continuation of phase I studies where the tolerated dose level appropriate for desired therapeutic action has been decided. This phase plays a vital role for prolongation of drug development into later phases. Depending on the tolerability, the new intervention can be determined to be included in other phases or not.

These are dose-ranging studies. The study end points and the nature of the population (severity of the disease) should be prefixed. Because the MTD is analyzed in the early phase, dose ranging is fixed among groups as placebo controlled, low, medium, and high (Fig. 23.12).

Certainly, the response to a defined dose can be studied (Fig. 23.13).

The dose-dependent pharmacological activity of the drug (mechanism of action) can be established. Therefore the dosing intervals and frequency of the drug administration can also be predicted for later phases. The therapeutic dose and its toxicity, if any, can be ascertained. The toxicity profile among the diseased population and their concomitant drugs provide a pathway for the continuation of phase 2b pivotal studies.

Phase 2b

These are pivotal studies, and new intervention may be compared with a dummy drug (placebo) or with an intervention already in existence. Thus these can be termed randomized controlled trials (RCTs) (Fig. 23.14).[20]

These are crossover studies and can be performed at multiple sites. As the dose requirements are determined in phase 2a, the efficacy of the dose is established (how well the suggested dose works on receptor mechanisms). The efficacy that has been proved should be determined from the target condition (a drug intended for a specific condition) as these are patient populations.

Phase 3—Therapeutic Confirmatory Phase

In a phase 3 trial, efficacy and safety of the molecule can be determined. These are therapeutic confirmatory phase trials. They are generally carried out in a relatively high number (100−3000) of patients, depending on therapeutic areas, at various centers, hence they are called multicentric trials. These trials are generally carried out for a longer duration (few years) to analyze the exact dosage levels that would prove beneficial to a lesser degree/extent of adverse events. With phase 3 study efficacy, postmarketing studies (phase 4) could study effectiveness. The difference between efficacy and effectiveness is compared in Table 23.6.

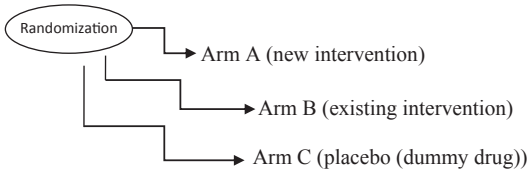


FIGURE 23.14 Randomized controlled trials—phase 2b.

TABLE 23.6 Difference Between Efficacy and Effectiveness

Efficacy	Effectiveness
Results from normal clinical trials (randomized)	Results from normal clinical practice
Efficacy is established in inclusion of population as specified in protocol	Effectiveness is established in inclusion of population as specified in protocol as well as with comorbidity conditions

A new intervention and a standard intervention are randomized among the patients. Sometimes a placebo-controlled trial may also be carried out. The researchers prefer to have a comparison, which could be more than two treatment groups. There are possibilities to study drug−drug interactions due to heterogeneous population selections.[21]

Example

Phase 3 Type 1 Diabetic Study

The possible occurrence of withdrawal cases due to adverse drug reactions could be more than expected. Assuming the clinical end point for a diabetic study would be lowering of (Fig. 23.15):

- $HbA1_C$ (glycosylated hemoglobin),
- fasting blood sugar (FBS),
- postprandial blood sugar,
- triglycerides and cholesterol.

The outcome of this study can be determined from the efficacy of the drug (mechanism of action of the drug on the receptor site, dose−response relationship) and its safety. The data should be statistically significant for expansion of the drug into the marketing phase. These are further categorized as phase 3a and phase 3b studies. *Phase 3a*: These studies are conducted after demonstrating the efficacy but before the submission of a new drug application (NDA). *Phase 3b*: These trials are carried out after NDA but before approval of the drug.

Phase 4—Postmarketing Surveillance

Phase 4 trials are conducted after a drug is approved and comes into a market. These trials will evaluate the effectiveness and safety of the drug in different population groups, not only in patients with intended diseases, but the drug will also be administered to patients with other concurrent diseases in real-world medical practice. Drugs during this phase, which show adverse toxicities, may not further be continued for marketing and could be banned in a given market. These trials are carried out in a patient undergoing clinical care. The statistical effectiveness of a drug on larger patient groups is known to a greater extent, and drug interactions can also be estimated as patients with a secondary illness may participate in the trials.[22]

Sometimes regulatory agencies may also request to conduct long-term postmarketing surveillance (PMS) studies to rule out the long-term adverse events, carcinogenicity, and mutagenic effects of the marketed drug. Data collected after such long-term PMS studies may describe potential long-term adverse reactions in which the drug may be withdrawn from the market.[22]

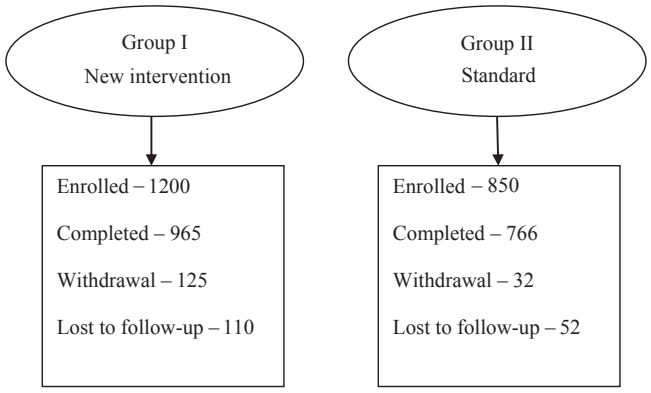


FIGURE 23.15 Example of participant withdrawal from clinical trial.

OBJECTIVE 23.4. DISCUSS THE ROLE OF PHARMACISTS AND NEWER TECHNOLOGIES IN CLINICAL RESEARCH

Pharmacists play important roles in the field of clinical research, and their roles could be:

1. Review of protocols and related documents,
2. Receipt, storage, accountability, dispensing, and return/destruction of IPs, and
3. Blinding/unblinding procedure of codes.

Review of Protocol and Trial-Related Documents

Clinical trial protocols should be prepared or reviewed by the pharmacist to analyze the following:

- The IP vs. control
- Mode of administration
- Dosage form and the dose required for each subject in each period
- Reconstitution solution, if any, in case of injectable
- Indications and uses of the IP
- Adverse effects of an IP
- Contraindications
- Preplanned concomitant medication, if any, to be administered in case of occurrence of adverse events

Pharmacists need to prepare and review protocols as per the pharmacological profile of study drugs. Pharmacists need to make sure that the objectives of a clinical research protocol comply with the pharmacology of the drug and the literature reporting scientific information.

Receipt, Storage, and Accountability of the Investigational Product

A pharmacist has important responsibilities in handling IPs. The number of IPs received, the date of manufacture and expiration, batch/lot numbers of the IPs, and route of drug administration should be accurately checked and documented. The handling and storage conditions (such as humidity, light, and temperature sensitivity) of the IPs as depicted on the label shall be maintained and followed.[32]

Drugs that need special storage conditions, such as narcotic substances or psychotropic substances, shall be preserved in a cupboard or safe, and the same shall be equipped with a security alarm system. The room in which the narcotics are placed should have a controlled access.[33]

Dispensing of the Investigational Product

A randomization schedule generated by the statistician should be used while dispensing. The dosage form and dose required for an individual should be cross-verified against the protocol before dispensing. Dispensing under specific requirements, if any (such as monochromatic light for light liable substances), is also essential for protocol compliance.

The key factors to be observed while dispensing are:

- Unintentional change of IP (test vs. control) that may occur due to overlooking of a randomization schedule to be avoided. Doing so may lead to the withdrawal of subjects from the trial.
- Dispensing extra IPs for the site(s). In case of any damage to packaging, which may be unnoticed during accountability, may once again this will lead to discontinuation of the trial. In some cases, spillage of the drug during intake may also occur.
- The accountability of unused IPs. The amount received should balance with dispensed and undispensed products.[32]

Return/Destruction of Investigational Products

The IPs may be returned to the sponsor after completion of the trial as per their request. Otherwise, the samples may be retained or destroyed as specified in the protocol and applicable regulatory guidelines.[34]

Blinding/Unblinding Procedure of Codes

Blinding of IP is essential in clinical trials for an unbiased result. The blinding procedure should be carried out by a pharmacist. The IPs are coded (i.e., test product is coded "A," and the control product is coded "B"). The blinded codes against IPs are kept in a sealed envelope under controlled access in the pharmacy. The randomization is then generated by the statistician using the blinded codes A or B.[34]

Dispensing the drugs should be performed by a pharmacist and sent across the sites (if multicentric) depending on the code generated against the number of subjects. Any labels for identification of the IP should be made based on the blinded codes only. The unused IP may be returned to the pharmacist.

Unblinding may happen for a subject(s) in case of a medical emergency condition. In such conditions, unblinding may be requested by the investigator to the pharmacist. The details of the subject and the reason for unblinding should be mentioned by the respective on-site investigator. The same should be notified to the sponsor at the earliest.[35]

In case of nonavailability of the pharmacist in such emergency conditions, other team members should be allocated for accessing and breaking the codes. Both blinding and unblinding should be documented by the pharmacist.

Details of the Pharmaceutical Development Used in Clinical Trials

Classical experimental designs were extended to clinical trials. However, before human responses to medical treatments are evaluated, the goal of product development is to design and establish a formulation composition and robust manufacturing process.

Quality by design for pharmaceuticals was introduced with ICH Q8 2009 (International Conference on Harmonization). There is a harmonized pharmaceutical quality system applicable across the lifecycle of the product that is developed following the guidelines:

- Q8—Pharmaceutical Development,
- Q2 (R2)—Pharmaceutical Development Revision,
- Q9—Quality Risk Management,
- Q10—Pharmaceutical Quality System,
- Q11—Development and Manufacture of Drug Substances (chemical/biological entities).[36]

The design of experiments (DOE) is the design of any task that describes or explains the variation of information under conditions that are hypothesized to reflect the variation. Experimental design implies not only the selection of suitable predictors (independent variables) and outcomes (dependent variables), but also planning the delivery of the experiment under statistically optimal conditions given the constraints of available resources. DOE involves making a set of experiments representative with regards to a given question, such as, for example, "What is the value of addition time for a controlled crystallization"—detection of short addition time could occur too late to prevent uncontrolled crystallization and thus impact over particle size distribution. Statistical analysis of crystallization data allows for the determination of the design space.

Researchers should select the system response, factors and their levels, and center point of the design (or the null point). Factors should be controllable and of high measurement precision. Factors could be temperature, agitator tip speed, concentration, and the time during the experiment, and the response could be the particle size.

Analysis of DOE data generates a predictive model:

$$y = \beta_0 + \sum_{i=1}^{k} \beta_i Xi + \sum_{i,j=1}^{k} \beta_{ij} XiXj + \sum_{i=1}^{k} \beta_{ii} Xi^2 + \ldots$$

where y is a response (predicted-calculated response value), aim function; Xi is the controllable independent variables, factors; β_0, β_i, β_{ij}, β_{ii} are real regression coefficients; and k is the number of factors. The model can be used to create a design space using narrower ranges and then used in the DOE.

A designed or active experiment is based on using general methodological concepts such as regression and correlation analysis, analysis of variance, randomization, optimal use of factor space, successive experimenting, replication, compactness of information, statistical estimates, etc.[19] It is very important to understand that a model is an approximation that simplifies studies of the reality. A model can never be 100% perfect, but it will be very useful in estimating interactions between factors. It is also very useful to inspect a reliable response contour plot of the investigated system to comprehend its behavior.

CONCLUSION

Key methodological aspects of clinical trials are explained in this chapter. Types of clinical trials and related studies are also discussed. Knowing these details would be beneficial to pharmacists in conducting research as well as to interpret the findings while reading the research.